

Complex Analysis Notes: Toward Euler's Formula

Contents

1	Complex Numbers	1
2	Algebraic Operations	2
3	Conjugates and Modulus	2
4	Reciprocals and Division	4
5	Complex Sequences	5
6	Complex Series	8
7	The Complex Exponential and Euler's Formula	11
8	Argument, Polar Form, and De Moivre's Theorem	16

1 Complex Numbers

Definition 1.1 (The complex numbers). *Define the complex numbers by*

$$\mathbb{C} = \mathbb{R}^2 \text{ with the usual vector addition and real scalar multiplication.}$$

Equip this real vector space with multiplication

$$(a, b)(c, d) = (ac - bd, ad + bc), \quad 1 = (1, 0), \quad i = (0, 1).$$

With this multiplication,

$$i^2 = (0, 1)(0, 1) = (-1, 0).$$

After identifying $a \in \mathbb{R}$ with $(a, 0) \in \mathbb{C}$, this says $i^2 = -1$. Every element $(a, b) \in \mathbb{R}^2$ can be written as

$$(a, b) = a(1, 0) + b(0, 1) = a + bi.$$

Definition 1.2 (Real and imaginary parts). *If $z = a + bi \in \mathbb{C}$, define*

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

Two complex numbers are equal precisely when their real and imaginary parts are equal:

$$a + bi = c + di \iff a = c \text{ and } b = d.$$

2 Algebraic Operations

Proposition 2.1 (Complex addition in $a + bi$ notation). *For $z = a + bi$ and $w = c + di$,*

$$z + w = (a + c) + (b + d)i.$$

Proof. This is the usual vector addition on $\mathbb{R}^2$ written in the notation $(a, b) = a + bi$. □

Proposition 2.2 (Complex multiplication in $a + bi$ notation). *For $z = a + bi$ and $w = c + di$,*

$$zw = (ac - bd) + (ad + bc)i.$$

Proof. This is exactly the multiplication

$$(a, b)(c, d) = (ac - bd, ad + bc)$$

written in the notation $(a, b) = a + bi$. □

Proposition 2.3 (Distributive law). *For all $z, w, u \in \mathbb{C}$,*

$$z(w + u) = zw + zu.$$

Proof. Let $z = a + bi$, $w = c + di$, and $u = e + fi$. Then

$$w + u = (c + e) + (d + f)i.$$

Hence

$$\begin{aligned} z(w + u) &= (a + bi)((c + e) + (d + f)i) \\ &= (a(c + e) - b(d + f)) + (a(d + f) + b(c + e))i \\ &= (ac - bd) + (ad + bc)i + (ae - bf) + (af + be)i \\ &= zw + zu. \end{aligned}$$

□

Definition 2.1 (Complex subtraction). *For $z, w \in \mathbb{C}$, define*

$$z - w := z + (-w).$$

If $z = a + bi$ and $w = c + di$, then

$$z - w = (a - c) + (b - d)i.$$

3 Conjugates and Modulus

Definition 3.1 (Complex conjugate). *If $z = a + bi$, define the complex conjugate of z by*

$$\bar{z} := a - bi.$$

Proposition 3.1 (Basic conjugate identities). *For all $z, w \in \mathbb{C}$,*

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w}, \quad \overline{\bar{z}} = z.$$

Proof. Each identity follows by writing $z = a + bi$ and $w = c + di$ and comparing real and imaginary parts. $\square$

Proposition 3.2 (Real part from conjugation). *For every $z \in \mathbb{C}$,*

$$z + \bar{z} = 2 \operatorname{Re}(z).$$

Proof. If $z = a + bi$, then

$$z + \bar{z} = (a + bi) + (a - bi) = 2a = 2 \operatorname{Re}(z).$$

$\square$

Definition 3.2 (Complex modulus). *If $z = a + bi$, define the modulus of z by*

$$|z| := \sqrt{a^2 + b^2}.$$

Proposition 3.3 (Modulus squared). *For every $z \in \mathbb{C}$,*

$$|z|^2 = z\bar{z}.$$

Proof. If $z = a + bi$, then

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2.$$

$\square$

Proposition 3.4 (Product rule for modulus). *For all $z, w \in \mathbb{C}$,*

$$|zw| = |z| |w|.$$

Proof. Using the conjugate product identity,

$$|zw|^2 = zw \overline{zw} = zw \bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2 |w|^2.$$

Both sides are nonnegative, so taking square roots gives $|zw| = |z| |w|$. $\square$

Proposition 3.5 (Powers and modulus). *For every $z \in \mathbb{C}$ and $n \in \mathbb{N}$,*

$$|z^n| = |z|^n.$$

Proof. This follows by induction from the product rule:

$$|z^{n+1}| = |z^n z| = |z^n| |z| = |z|^{n+1}.$$

$\square$

Proposition 3.6 (Real and imaginary parts are bounded by the modulus). *If $z \in \mathbb{C}$, then*

$$|\operatorname{Re}(z)| \leq |z|, \quad |\operatorname{Im}(z)| \leq |z|.$$

Proof. If $z = a + bi$, then $a^2 \leq a^2 + b^2$ and $b^2 \leq a^2 + b^2$. Taking square roots gives the result. $\square$

Proposition 3.7 (Triangle inequality). *For all $z, w \in \mathbb{C}$,*

$$|z + w| \leq |z| + |w|.$$

Proof. Since $\operatorname{Re}(\alpha) \leq |\alpha|$ for every $\alpha \in \mathbb{C}$,

$$\begin{aligned} |z + w|^2 &= (z + w)(\bar{z} + \bar{w}) \\ &= |z|^2 + |w|^2 + z\bar{w} + w\bar{z} \\ &= |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}) \\ &\leq |z|^2 + |w|^2 + 2|z\bar{w}| \\ &= |z|^2 + |w|^2 + 2|z||w| \\ &= (|z| + |w|)^2. \end{aligned}$$

Taking square roots gives the triangle inequality. □

Proposition 3.8 (Reverse triangle inequality). *For all $z, w \in \mathbb{C}$,*

$$||z| - |w|| \leq |z - w|.$$

Proof. By the triangle inequality,

$$|z| = |(z - w) + w| \leq |z - w| + |w|,$$

so $|z| - |w| \leq |z - w|$. Interchanging z and w gives

$$|w| - |z| \leq |w - z| = |z - w|.$$

Therefore $||z| - |w|| \leq |z - w|$. □

4 Reciprocals and Division

Definition 4.1 (Complex reciprocal). *If $z \in \mathbb{C}$ and $z \neq 0$, define*

$$\frac{1}{z} := \frac{\bar{z}}{|z|^2}.$$

Proposition 4.1 (The reciprocal is the multiplicative inverse). *If $z \neq 0$, then*

$$z \cdot \frac{1}{z} = 1.$$

Proof. Using $|z|^2 = z\bar{z}$,

$$z \cdot \frac{1}{z} = z \cdot \frac{\bar{z}}{|z|^2} = \frac{z\bar{z}}{|z|^2} = 1.$$

□

Proposition 4.2 ($\mathbb{C}$ is a field). *With the addition and multiplication defined above, $\mathbb{C}$ is a field.*

Proof. The additive identity is $0 = 0 + 0i$, and the multiplicative identity is $1 = 1 + 0i$. Associativity and commutativity follow by direct expansion from the corresponding real-number laws, and distributivity was proved above. If $z = a + bi$, then the additive inverse is

$$-z = (-a) + (-b)i.$$

If $z \neq 0$, then the reciprocal theorem gives a multiplicative inverse. Therefore $\mathbb{C}$ is a field. □

Definition 4.2 (Complex division). *For $w, z \in \mathbb{C}$ with $z \neq 0$, define*

$$\frac{w}{z} := w \cdot \frac{1}{z} = \frac{w\bar{z}}{|z|^2}.$$

5 Complex Sequences

Definition 5.1 (Complex sequence). *A complex sequence is a function*

$$z : \mathbb{N} \rightarrow \mathbb{C}.$$

We write z_n for $z(n)$. If $z_n = a_n + ib_n$, then (a_n) and (b_n) are real sequences, and

$$a_n = \operatorname{Re}(z_n), \quad b_n = \operatorname{Im}(z_n).$$

Definition 5.2 (ε - N convergence in $\mathbb{C}$). *Let (z_n) be a complex sequence and let $z \in \mathbb{C}$. We say*

$$z_n \rightarrow z$$

if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |z_n - z| < \varepsilon.$$

Proposition 5.1 (Convergence as distance to the limit). *Let (z_n) be a complex sequence and let $z \in \mathbb{C}$. Then*

$$z_n \rightarrow z \iff |z_n - z| \rightarrow 0$$

as a real sequence.

Proof. The sequence $(|z_n - z|)$ is a real sequence because each term is a nonnegative real number.

First suppose $z_n \rightarrow z$ in $\mathbb{C}$. By definition, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |z_n - z| < \varepsilon.$$

This is exactly the statement that the real sequence $|z_n - z|$ converges to 0.

Conversely, suppose $|z_n - z| \rightarrow 0$ as a real sequence. Then for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies ||z_n - z| - 0| < \varepsilon.$$

Since $|z_n - z| \geq 0$,

$$||z_n - z| - 0| = |z_n - z|.$$

Thus

$$n > N \implies |z_n - z| < \varepsilon,$$

which is precisely $z_n \rightarrow z$ in $\mathbb{C}$. □

Theorem 5.1 (Componentwise convergence). *Let $z_n = a_n + ib_n$ and $z = a + ib$. Then*

$$z_n \rightarrow z \iff a_n \rightarrow a \text{ and } b_n \rightarrow b.$$

Equivalently,

$$z_n \rightarrow z \iff \operatorname{Re}(z_n) \rightarrow \operatorname{Re}(z) \text{ and } \operatorname{Im}(z_n) \rightarrow \operatorname{Im}(z).$$

Proof. First suppose $z_n \rightarrow z$. Since

$$|a_n - a| = |\operatorname{Re}(z_n - z)| \leq |z_n - z|, \quad |b_n - b| = |\operatorname{Im}(z_n - z)| \leq |z_n - z|,$$

the ε - N definition for $z_n \rightarrow z$ implies $a_n \rightarrow a$ and $b_n \rightarrow b$.

Conversely, suppose $a_n \rightarrow a$ and $b_n \rightarrow b$. Let $\varepsilon > 0$. Choose N_1 such that

$$n > N_1 \implies |a_n - a| < \frac{\varepsilon}{2},$$

and choose N_2 such that

$$n > N_2 \implies |b_n - b| < \frac{\varepsilon}{2}.$$

Let $N = \max\{N_1, N_2\}$. Then for $n > N$,

$$\begin{aligned} |z_n - z| &= |(a_n - a) + i(b_n - b)| \\ &= \sqrt{(a_n - a)^2 + (b_n - b)^2} \\ &\leq |a_n - a| + |b_n - b| \\ &< \varepsilon. \end{aligned}$$

Therefore $z_n \rightarrow z$. □

Theorem 5.2 (Uniqueness of limits). *If $z_n \rightarrow z$ and $z_n \rightarrow w$, then $z = w$.*

Proof. Write

$$z = a + ib, \quad w = c + id, \quad z_n = a_n + ib_n.$$

By componentwise convergence,

$$a_n \rightarrow a, \quad b_n \rightarrow b, \quad a_n \rightarrow c, \quad b_n \rightarrow d.$$

Real limits are unique, so $a = c$ and $b = d$. Therefore $z = w$. □

Theorem 5.3 (Convergent sequences are bounded). *If $z_n \rightarrow z$, then (z_n) is bounded: there exists $M > 0$ such that*

$$|z_n| \leq M \quad \forall n \in \mathbb{N}.$$

Proof. Since $z_n \rightarrow z$, choose $N \in \mathbb{N}$ such that

$$n > N \implies |z_n - z| < 1.$$

For $n > N$, the triangle inequality gives

$$|z_n| = |(z_n - z) + z| \leq |z_n - z| + |z| < 1 + |z|.$$

Now define

$$M := \max\{|z_1|, |z_2|, \dots, |z_N|, 1 + |z|\}.$$

Then $|z_n| \leq M$ for every $n \in \mathbb{N}$. □

Definition 5.3 (Subsequence). *Let (z_n) be a complex sequence. A subsequence is a sequence of the form*

$$z_{n_k},$$

where

$$n_1 < n_2 < n_3 < \dots$$

is a strictly increasing sequence of natural numbers.

Proposition 5.2 (Subsequences preserve limits). *If $z_n \rightarrow z$, then every subsequence z_{n_k} also converges to z .*

Proof. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$n > N \implies |z_n - z| < \varepsilon.$$

Since $n_k \geq k$, every $k > N$ satisfies $n_k > N$. Thus

$$k > N \implies |z_{n_k} - z| < \varepsilon.$$

Therefore $z_{n_k} \rightarrow z$. □

Definition 5.4 (Cauchy sequence in $\mathbb{C}$). *A complex sequence (z_n) is Cauchy if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$m, n > N \implies |z_n - z_m| < \varepsilon.$$

Theorem 5.4 (Cauchy criterion in $\mathbb{C}$). *A complex sequence (z_n) converges if and only if it is Cauchy.*

Proof. Write $z_n = a_n + ib_n$.

Suppose $z_n \rightarrow z$. Every convergent sequence is Cauchy: given $\varepsilon > 0$, choose N such that

$$n > N \implies |z_n - z| < \frac{\varepsilon}{2}.$$

Then for $m, n > N$,

$$|z_n - z_m| \leq |z_n - z| + |z_m - z| < \varepsilon.$$

Conversely, suppose (z_n) is Cauchy. Since

$$|a_n - a_m| \leq |z_n - z_m|, \quad |b_n - b_m| \leq |z_n - z_m|,$$

the real sequences (a_n) and (b_n) are Cauchy. Because $\mathbb{R}$ is complete, there exist $a, b \in \mathbb{R}$ such that

$$a_n \rightarrow a, \quad b_n \rightarrow b.$$

By componentwise convergence, $z_n \rightarrow a + ib$. □

Theorem 5.5 (Limit algebra). *Suppose $z_n \rightarrow z$ and $w_n \rightarrow w$ in $\mathbb{C}$, and let $\alpha \in \mathbb{C}$. Then*

$$z_n + w_n \rightarrow z + w, \quad z_n - w_n \rightarrow z - w, \quad \alpha z_n \rightarrow \alpha z, \quad z_n w_n \rightarrow zw.$$

If $w \neq 0$ and $w_n \neq 0$ eventually, then

$$\frac{z_n}{w_n} \rightarrow \frac{z}{w}.$$

Proof. For sums,

$$|(z_n + w_n) - (z + w)| \leq |z_n - z| + |w_n - w| \rightarrow 0.$$

Subtraction follows from addition and scalar multiplication by -1 . For scalar multiplication,

$$|\alpha z_n - \alpha z| = |\alpha| |z_n - z| \rightarrow 0.$$

For products,

$$\begin{aligned} z_n w_n - z w &= z_n w_n - z_n w + z_n w - z w \\ &= z_n (w_n - w) + (z_n - z) w. \end{aligned}$$

Since (z_n) is convergent, it is bounded; choose $M > 0$ such that $|z_n| \leq M$ for all n . Then

$$|z_n w_n - z w| \leq M |w_n - w| + |w| |z_n - z| \rightarrow 0.$$

For reciprocals, assume $w \neq 0$. Since $w_n \rightarrow w$, choose N such that

$$n > N \implies |w_n - w| < \frac{|w|}{2}.$$

Then for $n > N$,

$$|w_n| \geq |w| - |w_n - w| > \frac{|w|}{2}.$$

Therefore

$$\left| \frac{1}{w_n} - \frac{1}{w} \right| = \left| \frac{w - w_n}{w_n w} \right| = \frac{|w_n - w|}{|w_n| |w|} \leq \frac{2}{|w|^2} |w_n - w| \rightarrow 0.$$

Thus $1/w_n \rightarrow 1/w$, and the quotient rule follows from the product rule:

$$\frac{z_n}{w_n} = z_n \cdot \frac{1}{w_n} \rightarrow z \cdot \frac{1}{w}.$$

□

Proposition 5.3 (Limit of the modulus). *If $z_n \rightarrow z$, then*

$$|z_n| \rightarrow |z|.$$

Proof. By the reverse triangle inequality,

$$||z_n| - |z|| \leq |z_n - z|.$$

Since $|z_n - z| \rightarrow 0$, it follows that $|z_n| \rightarrow |z|$. □

Proposition 5.4 (Limit of the conjugate). *If $z_n \rightarrow z$, then*

$$\overline{z_n} \rightarrow \overline{z}.$$

Proof. Since

$$|\overline{z_n} - \overline{z}| = |\overline{z_n - z}| = |z_n - z|,$$

we have $\overline{z_n} \rightarrow \overline{z}$. □

6 Complex Series

Definition 6.1 (Partial sum sequence). *Let $(z_n)_{n=0}^{\infty}$ be a complex sequence. The associated partial sum sequence $(s_N)_{N=0}^{\infty}$ is defined by*

$$s_N := \sum_{n=0}^N z_n = z_0 + z_1 + \cdots + z_N.$$

Definition 6.2 (Infinite sum). Let $(z_n)_{n=0}^{\infty}$ be a complex sequence with partial sums

$$s_N = \sum_{n=0}^N z_n.$$

If the partial sum sequence converges to some $S \in \mathbb{C}$, then the infinite series converges and we write

$$\sum_{n=0}^{\infty} z_n = S.$$

Equivalently,

$$\sum_{n=0}^{\infty} z_n = S \iff \lim_{N \rightarrow \infty} \sum_{n=0}^N z_n = S.$$

Definition 6.3 (Absolute convergence). A complex series $\sum_{n=0}^{\infty} z_n$ is absolutely convergent if the real series

$$\sum_{n=0}^{\infty} |z_n|$$

converges.

Theorem 6.1 (Absolute convergence implies convergence). If $\sum_{n=0}^{\infty} z_n$ is absolutely convergent, then $\sum_{n=0}^{\infty} z_n$ converges.

Proof. Let

$$s_N := \sum_{n=0}^N z_n, \quad t_N := \sum_{n=0}^N |z_n|.$$

Since $\sum |z_n|$ converges in $\mathbb{R}$, the real partial sums (t_N) are Cauchy. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$q > p > N \implies \sum_{n=p+1}^q |z_n| = t_q - t_p < \varepsilon.$$

Then for $q > p > N$,

$$|s_q - s_p| = \left| \sum_{n=p+1}^q z_n \right| \leq \sum_{n=p+1}^q |z_n| < \varepsilon.$$

Thus (s_N) is Cauchy in $\mathbb{C}$, so it converges by the Cauchy criterion in $\mathbb{C}$. Therefore $\sum z_n$ converges. $\square$

Theorem 6.2 (Reordering absolutely convergent series). Suppose $\sum_{n=0}^{\infty} z_n$ is absolutely convergent and

$$\sum_{n=0}^{\infty} z_n = S.$$

If $\sigma : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is a bijection, then

$$\sum_{n=0}^{\infty} z_{\sigma(n)}$$

also converges absolutely, and

$$\sum_{n=0}^{\infty} z_{\sigma(n)} = S.$$

Proof. First, the reordered absolute series converges. For every N , the finite set $\{\sigma(0), \dots, \sigma(N)\}$ is contained in $\{0, \dots, M\}$ for some M , so

$$\sum_{n=0}^N |z_{\sigma(n)}| \leq \sum_{n=0}^M |z_n| \leq \sum_{n=0}^{\infty} |z_n|.$$

Thus the partial sums of $\sum |z_{\sigma(n)}|$ are increasing and bounded, hence converge.

Now let

$$s_N := \sum_{n=0}^N z_n, \quad r_N := \sum_{n=0}^N z_{\sigma(n)}.$$

Let $\varepsilon > 0$. Since $\sum |z_n|$ converges and $s_K \rightarrow S$, choose K such that

$$\sum_{j=K+1}^{\infty} |z_j| < \varepsilon \quad \text{and} \quad |s_K - S| < \varepsilon.$$

Choose N large enough that

$$\{0, 1, \dots, K\} \subseteq \{\sigma(0), \sigma(1), \dots, \sigma(N)\}.$$

For $m > N$, the difference $r_m - s_K$ is a finite sum of terms z_j with $j > K$. Therefore

$$|r_m - s_K| \leq \sum_{\substack{j > K \\ j \in \{\sigma(0), \dots, \sigma(m)\}}} |z_j| \leq \sum_{j=K+1}^{\infty} |z_j| \leq \varepsilon.$$

Then for $m > N$,

$$|r_m - S| \leq |r_m - s_K| + |s_K - S| < 2\varepsilon.$$

Hence $r_m \rightarrow S$, so the reordered series converges to the same sum. $\square$

Theorem 6.3 (Ratio test). *Let $\sum_{n=0}^{\infty} z_n$ be a complex series, and suppose $z_n \neq 0$ for all sufficiently large n . If*

$$\lim_{n \rightarrow \infty} \frac{|z_{n+1}|}{|z_n|} = L,$$

then:

$$L < 1 \implies \sum_{n=0}^{\infty} z_n \text{ converges absolutely,}$$

$$L > 1 \implies \sum_{n=0}^{\infty} z_n \text{ diverges,}$$

$$L = 1 \implies \text{the test is inconclusive.}$$

Proof. First suppose $L < 1$. Choose r such that

$$L < r < 1.$$

Since $|z_{n+1}|/|z_n| \rightarrow L$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies \frac{|z_{n+1}|}{|z_n|} < r.$$

Thus, for $k \geq 1$,

$$|z_{N+k}| < r^k |z_N|.$$

Therefore

$$\sum_{k=1}^{\infty} |z_{N+k}| \leq |z_N| \sum_{k=1}^{\infty} r^k < \infty.$$

Adding finitely many initial terms does not affect convergence, so $\sum |z_n|$ converges. Hence $\sum z_n$ converges absolutely.

Now suppose $L > 1$. Choose $r > 1$ such that

$$1 < r < L.$$

Then for all sufficiently large n ,

$$\frac{|z_{n+1}|}{|z_n|} > r,$$

so $|z_{n+1}| > r |z_n|$ eventually. In particular, $|z_n|$ does not converge to 0. Hence z_n does not converge to 0, so the series $\sum z_n$ diverges.

When $L = 1$, the test gives no conclusion. For example, $\sum 1/n$ diverges but $\sum 1/n^2$ converges, and both have ratio limit 1. $\square$

7 The Complex Exponential and Euler's Formula

Theorem 7.1 (Binomial theorem over $\mathbb{C}$). *For all $z, w \in \mathbb{C}$ and $n \in \mathbb{N}$,*

$$(z + w)^n = \sum_{k=0}^n \binom{n}{k} z^k w^{n-k}.$$

Proof. The proof is the same induction as over $\mathbb{R}$, using that $\mathbb{C}$ is a commutative field. For $n = 1$, the statement is immediate. Assume it holds for n . Then

$$\begin{aligned} (z + w)^{n+1} &= (z + w) \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} z^{k+1} w^{n-k} + \sum_{k=0}^n \binom{n}{k} z^k w^{n+1-k}. \end{aligned}$$

Collecting equal powers and using Pascal's identity gives

$$(z + w)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} z^k w^{n+1-k}.$$

$\square$

Theorem 7.2 (Cauchy product theorem). *Suppose $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are absolutely convergent complex series, and define*

$$c_n := \sum_{k=0}^n a_k b_{n-k}.$$

Then $\sum c_n$ converges absolutely and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right).$$

Proof. Let

$$A_* := \sum_{n=0}^{\infty} |a_n|, \quad B_* := \sum_{n=0}^{\infty} |b_n|.$$

For every N ,

$$\sum_{n=0}^N |c_n| \leq \sum_{n=0}^N \sum_{k=0}^n |a_k| |b_{n-k}| \leq A_* B_*.$$

Thus $\sum c_n$ is absolutely convergent.

Let

$$A = \sum_{n=0}^{\infty} a_n, \quad B = \sum_{n=0}^{\infty} b_n,$$

and define

$$A_N := \sum_{n=0}^N a_n, \quad B_N := \sum_{n=0}^N b_n, \quad C_N := \sum_{n=0}^N c_n.$$

Then

$$A_N B_N = \sum_{\substack{0 \leq k \leq N \\ 0 \leq j \leq N}} a_k b_j, \quad C_N = \sum_{\substack{k, j \geq 0 \\ k+j \leq N}} a_k b_j.$$

Choose K so large that both tails

$$\sum_{k>K} |a_k|, \quad \sum_{j>K} |b_j|$$

are small. If $N > 2K$, then every pair satisfying $0 \leq k, j \leq N$ and $k+j > N$ has either $k > K$ or $j > K$. Hence

$$|A_N B_N - C_N| \leq B_* \sum_{k>K} |a_k| + A_* \sum_{j>K} |b_j|,$$

which can be made arbitrarily small. Since $A_N \rightarrow A$ and $B_N \rightarrow B$, we have $A_N B_N \rightarrow AB$. Therefore $C_N \rightarrow AB$. $\square$

Definition 7.1 (The complex exponential). For $z \in \mathbb{C}$, define

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Proposition 7.1 (The exponential series converges absolutely). For every $z \in \mathbb{C}$, the series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

converges absolutely.

Proof. If $z = 0$, then

$$\sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + 0 + 0 + \dots$$

is absolutely convergent.

Now suppose $z \neq 0$. Let

$$a_n := \frac{z^n}{n!}.$$

Then $a_n \neq 0$ for every n , and

$$\frac{|a_{n+1}|}{|a_n|} = \frac{|z^{n+1}|}{(n+1)!} \frac{n!}{|z^n|} = \frac{|z|}{n+1} \rightarrow 0.$$

By the ratio test, $\sum a_n$ converges absolutely. □

Theorem 7.3 (Exponential addition law). *For all $z, w \in \mathbb{C}$,*

$$\exp(z + w) = \exp(z) \exp(w).$$

Proof. The exponential series for z and w converge absolutely, so their product is given by the Cauchy product:

$$\begin{aligned} \exp(z) \exp(w) &= \left(\sum_{n=0}^{\infty} \frac{z^n}{n!} \right) \left(\sum_{n=0}^{\infty} \frac{w^n}{n!} \right) \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{z^k}{k!} \frac{w^{n-k}}{(n-k)!}. \end{aligned}$$

Since

$$\frac{1}{k!(n-k)!} = \frac{1}{n!} \binom{n}{k},$$

we get

$$\begin{aligned} \exp(z) \exp(w) &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \\ &= \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} \\ &= \exp(z+w), \end{aligned}$$

where the binomial theorem was used in the second-to-last line. □

Definition 7.2 (Sine and cosine by power series). *For $z \in \mathbb{C}$, define*

$$\cos z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \sin z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

Proposition 7.2 (The sine and cosine series converge absolutely). *For every $z \in \mathbb{C}$, the series defining $\cos z$ and $\sin z$ converge absolutely.*

Proof. If $z = 0$, convergence is immediate. Assume $z \neq 0$. For the cosine series, let

$$a_n := (-1)^n \frac{z^{2n}}{(2n)!}.$$

Then

$$\frac{|a_{n+1}|}{|a_n|} = \frac{|z|^2}{(2n+2)(2n+1)} \rightarrow 0.$$

For the sine series, let

$$b_n := (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

Then

$$\frac{|b_{n+1}|}{|b_n|} = \frac{|z|^2}{(2n+3)(2n+2)} \rightarrow 0.$$

Both series converge absolutely by the ratio test. □

Theorem 7.4 (Euler's formula over $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$\exp(iz) = \cos z + i \sin z.$$

Proof. By definition,

$$\exp(iz) = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!}.$$

This exponential series is absolutely convergent, so its terms may be reordered and grouped without changing the sum. Separate the even and odd powers:

$$\exp(iz) = \sum_{n=0}^{\infty} \frac{(iz)^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{(iz)^{2n+1}}{(2n+1)!}.$$

Since

$$i^{2n} = (-1)^n, \quad i^{2n+1} = i(-1)^n,$$

we get

$$\begin{aligned} \exp(iz) &= \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} + i \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \\ &= \cos z + i \sin z. \end{aligned}$$

□

Theorem 7.5 (Exponential formulas for sine and cosine). *For every $z \in \mathbb{C}$,*

$$\boxed{\cos z = \frac{\exp(iz) + \exp(-iz)}{2}}, \quad \boxed{\sin z = \frac{\exp(iz) - \exp(-iz)}{2i}}.$$

Proof. From the power-series definitions,

$$\cos(-z) = \cos z, \quad \sin(-z) = -\sin z.$$

Euler's formula applied to z and $-z$ gives

$$\exp(iz) = \cos z + i \sin z, \quad \exp(-iz) = \cos z - i \sin z.$$

Adding these identities gives

$$\exp(iz) + \exp(-iz) = 2 \cos z.$$

Subtracting gives

$$\exp(iz) - \exp(-iz) = 2i \sin z.$$

□

Theorem 7.6 (Euler's identity for real angles). *For every $\theta \in \mathbb{R}$,*

$$\exp(i\theta) = \cos \theta + i \sin \theta.$$

Proof. This is the special case of Euler's formula over $\mathbb{C}$ with $z = \theta \in \mathbb{R}$. □

Theorem 7.7 (Real and imaginary parts of $\exp z$). *Let $z = a + ib$ with $a, b \in \mathbb{R}$. Then*

$$\exp(z) = \exp(a)(\cos b + i \sin b).$$

Consequently,

$$\operatorname{Re}(\exp z) = \exp(a) \cos b, \quad \operatorname{Im}(\exp z) = \exp(a) \sin b.$$

Equivalently,

$$\operatorname{Re}(\exp z) = \exp(\operatorname{Re} z) \cos(\operatorname{Im} z), \quad \operatorname{Im}(\exp z) = \exp(\operatorname{Re} z) \sin(\operatorname{Im} z).$$

Proof. Using the exponential addition law and Euler's identity for real angles,

$$\exp(z) = \exp(a + ib) = \exp(a) \exp(ib) = \exp(a)(\cos b + i \sin b).$$

Since $a, b \in \mathbb{R}$, the power-series definitions give $\exp(a), \cos b, \sin b \in \mathbb{R}$, so

$$\exp(z) = \exp(a) \cos b + i \exp(a) \sin b.$$

Reading off real and imaginary parts gives the result. □

Theorem 7.8 (Modulus of the complex exponential). *For every $z \in \mathbb{C}$,*

$$|\exp(z)| = \exp(\operatorname{Re} z).$$

In particular, for every $\theta \in \mathbb{R}$,

$$|\exp(i\theta)| = 1.$$

Proof. Write $z = a + ib$ with $a, b \in \mathbb{R}$. From the previous theorem,

$$\exp(z) = \exp(a)(\cos b + i \sin b).$$

Thus

$$\begin{aligned} |\exp(z)| &= |\exp(a)| |\cos b + i \sin b| \\ &= \exp(a) \sqrt{\cos^2 b + \sin^2 b} \\ &= \exp(a). \end{aligned}$$

Here we use the standard real-analysis facts that $\exp(a) > 0$ for real a and $\cos^2 b + \sin^2 b = 1$. Since $a = \operatorname{Re} z$, this proves $|\exp(z)| = \exp(\operatorname{Re} z)$. If $z = i\theta$, then $\operatorname{Re}(i\theta) = 0$, so

$$|\exp(i\theta)| = \exp(0) = 1.$$

□

8 Argument, Polar Form, and De Moivre's Theorem

Definition 8.1 (Argument). Let $z \in \mathbb{C}$ with $z \neq 0$. An argument of z is any real number θ such that

$$z = |z|(\cos \theta + i \sin \theta).$$

The set of all arguments of z is denoted

$$\arg(z).$$

Arguments are not unique: if $\theta \in \arg(z)$, then

$$\arg(z) = \{\theta + 2\pi k : k \in \mathbb{Z}\}.$$

Definition 8.2 (Polar form). If $z \neq 0$ and $\theta \in \arg(z)$, then the polar form of z is

$$z = r(\cos \theta + i \sin \theta), \quad r = |z| > 0.$$

Using Euler's formula, this can also be written as

$$z = re^{i\theta}.$$

Proposition 8.1 (Existence of polar form). Every nonzero complex number has a polar form.

Proof. Let $z = a + ib \neq 0$ and set $r = |z| = \sqrt{a^2 + b^2}$. Then $r > 0$, and

$$\left(\frac{a}{r}\right)^2 + \left(\frac{b}{r}\right)^2 = 1.$$

By the standard real parametrization of the unit circle, there exists $\theta \in \mathbb{R}$ such that

$$\cos \theta = \frac{a}{r}, \quad \sin \theta = \frac{b}{r}.$$

Therefore

$$z = a + ib = r(\cos \theta + i \sin \theta) = re^{i\theta}.$$

□

Proposition 8.2 (Multiplication in polar form). If

$$z = re^{i\theta}, \quad w = se^{i\phi},$$

with $r, s > 0$, then

$$zw = rs e^{i(\theta+\phi)}.$$

Proof. Using the exponential addition law,

$$zw = (re^{i\theta})(se^{i\phi}) = rs e^{i\theta} e^{i\phi} = rs e^{i(\theta+\phi)}.$$

□

Theorem 8.1 (De Moivre's theorem). For every $\theta \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Equivalently,

$$(e^{i\theta})^n = e^{in\theta}.$$

Proof. By Euler's formula,

$$\cos \theta + i \sin \theta = e^{i\theta}.$$

Using the exponential addition law repeatedly,

$$(e^{i\theta})^n = \underbrace{e^{i\theta} \cdots e^{i\theta}}_{n \text{ factors}} = e^{i(\theta + \cdots + \theta)} = e^{in\theta}.$$

Applying Euler's formula again gives

$$e^{in\theta} = \cos(n\theta) + i \sin(n\theta).$$

□

Useful Real and Imaginary Parts

Expression	$\operatorname{Re}(\cdot)$	$\operatorname{Im}(\cdot)$
$z = a + bi$	a	b
$\bar{z} = a - bi$	a	$-b$
$z + w, w = c + di$	$a + c$	$b + d$
$zw, w = c + di$	$ac - bd$	$ad + bc$
$z^{-1}, z \neq 0$	$\frac{a}{a^2 + b^2}$	$-\frac{b}{a^2 + b^2}$
$\frac{w}{z}, z \neq 0$	$\frac{ac + bd}{a^2 + b^2}$	$\frac{ad - bc}{a^2 + b^2}$
$\exp(a + bi)$	$\exp(a) \cos b$	$\exp(a) \sin b$
$\exp(i\theta)$	$\cos \theta$	$\sin \theta$
$re^{i\theta}, r > 0$	$r \cos \theta$	$r \sin \theta$